

AE1601 Glider Project Report

Fall 2024 | Section A | Group 14

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Introduction

- Create a glider primarily from balsa wood
- Glider must fly a minimum of 20 feet
- Consider pros/cons of different design alternations
- Create design iterations to increase measures of success
- Use principles learned from class to help construct glider
- May the furthest glider win!

Requirements

- Plan and create a glider from primarily balsa wood
- Conduct at least one design iteration to the initial design
- Final glider must fit within a 16 in x 16 in box
- Must be a glider (contain wings & no self-contained propulsion)
- Must be hand-thrown
- Must remain intact after throw
- Must fly at least 20 feet

Measures of Success for Requirements

- **Design Measures of Success**

- Checkpoint 1: Initial design
- Checkpoint 2: Testing and iterations

- **Construction Requirements Measures of Success**

- Material Performance
- Wingshape & Size
- Balance & Weight (CG)

- **Flight Requirements Measures of Success**

- Distance & Intactness

Design Alternatives & Pros/Cons - Wingshape

- **Rectangle**

- Easy to make & High wing area
- Good stability
- High drag

- **Tapered**

- High lift & Reduced drag
- Susceptible to tip stall

- **Delta**

- High lift & Low drag
- Good stability
- High angle of attack

- **Elliptical**

- Efficient & Stable
- Reduced wingtip vortex
- Hard to construct

- **Sweptback**

- Drag reduction
- Not ideal for a glider
- Efficient at high speeds

Design Alternatives & Pros/Cons - Material

- **Balsawood**

- Requires laser cutter for utmost precision
- Longevity
- Scarce quantity

- **Cardboard**

- Easy to cut and construct
- Lacks structural integrity
- Abundant quantity

- **3D-Print Material**

- Not within requirements
- Long processing time

Design Alternatives & Pros/Cons - Adhesive

- **Masking Tape**

- Abundant
- Can be misleading in terms of results
- Bulky

- **Hot Glue**

- Quick application
- Bulky
- Can affect aerodynamics if applied unevenly

- **Super Glue**

- Quick application
- Lightweight
- Strong hold

- **CA**

- Hard to find/access
- Lightweight
- Very strong hold

Highest Probability of Success Design

- **Wingshape**

- Testing: Tapered, Sweptback, Delta
- Final Build: Delta
 - Delta wing flew further than the other wingshape designs, and it was easier to plan/make

- **Material**

- Testing: Cardboard, Balsa-Cardboard Mix
- Final Build: Balsa
 - Balsa was both lightweight and sturdy, and had longevity/endurance

- **Adhesive**

- Testing: Masking Tape, Super Glue
- Final Build: Super Glue
 - Strong hold and had no aerodynamic affect – would have used CA if acquired

Sizing

- Fuselage Length: 10 inches
- Wingspan: 12 inches
 - Tip Chord: 1 inch
 - Root Chord: 2 inches
 - Location: 2 inches from nose
- Horizontal Tail Span: 5 inches
 - Horizontal Tail Root Chord: 1.5 inches
 - Horizontal Tail Tip Chord: 0.75 inches
- Vertical Tail Base: 1.5 inches
- Vertical Tail Height: 3 inches

Initial Design Choices

- **Tapered Wing**
 - Simplicity
 - Smooth construction
- **Prototyping**
 - Cardboard material
 - Modular wing mount
 - Excessive masking tape



Testing (Tapered Wing)

- **Stability**

- Tapered prototype flew fairly straight
 - Inconsistency likely due to lack of sturdiness
- Flipped forward due to lack of forefront mass

- **Initial Performance versus Requirements**

- Initially did not travel past 10 ft
 - Once mass (play-doh) was added to nose, distance increased significantly
- Post addition of mass - average distance ~25 ft
 - This met minimum requirements but was still unsatisfactory

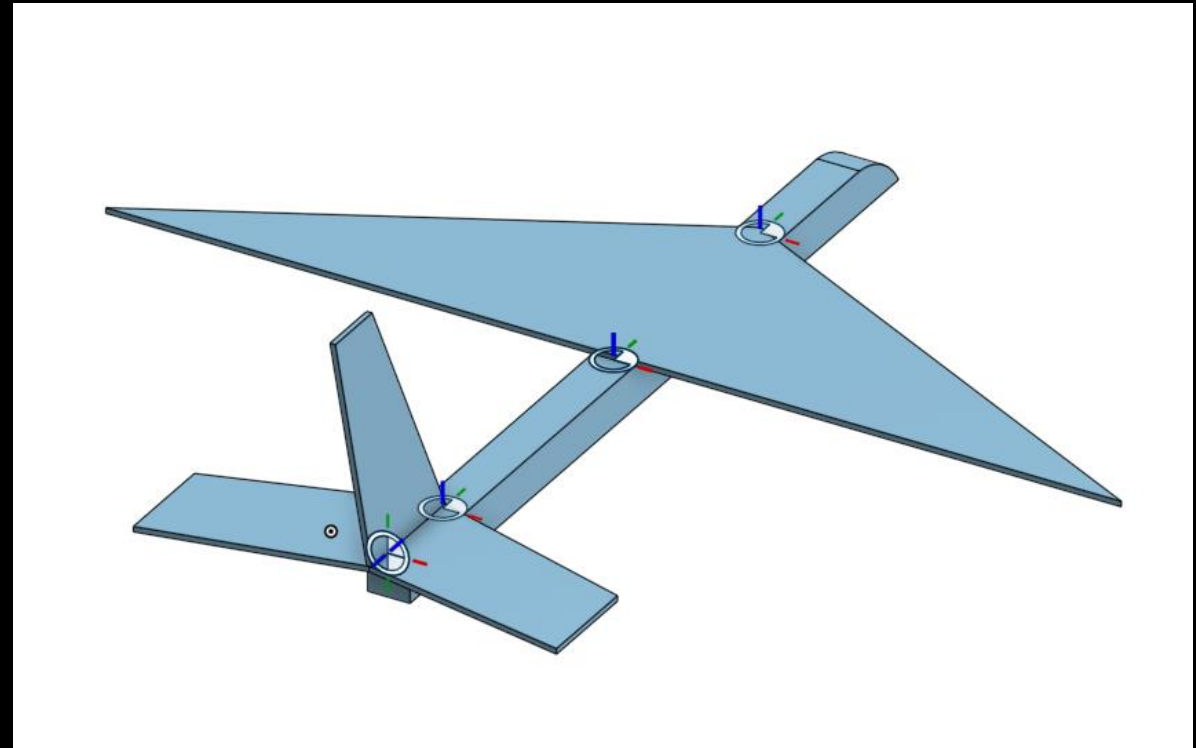
Design Iteration

- **Aspect:**

- Changing wingshape from tapered to delta
- More forceful throws

- **Planned Changes:**

- Wingspan to be 12 inches
 - Not too long; not too short
- Root cord to be 3 inches
 - Max length due to given balsa sheet
- Increased structure/stability
 - Intactness upon "touchdown"
- Higher angle of attack



Testing (Delta Wing)

- **Stability**

- Greatly increased stability
 - Close to perfect flight path
 - No wobbling

- **Initial Performance versus Requirements**

- Initially traveled 40 ft
 - Extreme increase from tapered
- Post addition of mass - average distance ~75 ft
 - This exceeded the 20 ft requirement over three-fold



Final Design

- Delta Wing
- Sweptback Horizontal Stabilizer
- Sweptback Vertical Stabilizer
- Mass component (Play-Doh) of 25 g
- Consistent ~75 ft throws
- Total Mass: 35 grams



Calculations – Center of Gravity

- 2 inches away from the nose
- 0.5 inches in front of the $\frac{1}{4}$ chord
- Indicated with red dot on figure



Calculations – Coefficient of Lift

$$m = 35 \text{ grams}$$

$$W = 0.0772 \text{ lbs}$$

$$S = (0.5)(1 \text{ ft})(0.25 \text{ ft}) = 0.125 \text{ ft}^2$$

$$b = 1 \text{ ft}$$

$$T = 2.9 \text{ sec}$$

$$d = 75 \text{ ft}$$

$$V = 75 \text{ ft} / 2.9 \text{ sec} = 25.862 \text{ ft/sec}$$

$$\rho = 0.0765 \text{ lb/ft}^3$$

$$C_L = \frac{2W}{\rho V^2 S} = \frac{2(0.0772 \text{ lbs})}{\left(0.0765 \frac{\text{lbs}}{\text{ft}^3}\right) \left(25.862 \frac{\text{ft}}{\text{sec}}\right)^2 (0.125 \text{ ft}^2)}$$

$$C_L = 0.0241$$

Calculations – Coefficient of Drag

$$C_L = 0.0241$$

$$e = 0.90$$

$$b = 1 \text{ ft}$$

$$S = 0.125 \text{ ft}^2$$

$$AR = 8$$

$$C_{D_i} = \frac{C_L^2}{\pi e AR} = \frac{(0.0241)^2}{(\pi)(0.90)(8)}$$

$$C_{D_i} = 0.0000257$$

$$C_D = 0.0000514$$

Design Iteration on Final Performance

- **Increased Flight Distance**

- ~25 ft to ~75 ft
- More harder/forceful throws
- Furthest throw during competition: 79 feet

- **Increased Stability**

- Straight flight path
- No teetering
- Stable yaw & pitch

Best Test Flight

- **Video:** 64 ft
- **Best Ever:** 106 ft
(unrecorded)



Final Design versus Requirements

- Plan and create a glider from primarily balsa wood ✓
- Conduct at least one design iteration to the initial design ✓
- Final glider must fit within a 16 in x 16 in box ✓
- Glider must consist of primarily balsa wood ✓
- Must be a glider (contain wings & no self-contained propulsion) ✓
- Must be hand-thrown ✓
- Must remain intact after throw ✓
- Must fly at least 20 feet ✓

Conclusion – Summary

- Section A – 2nd place 🏆 (79 feet)
- Key Aspects to a Successful Build:
 - Planning
 - Researching
 - Prototyping
 - Iterating
 - Collaboration
- Application of aerodynamic principles learned from lecture
- Overall project was a SUCCESS!

Conclusion – Further Improvements

- **Higher Aspect Ratio**

- Increase wingspan
- Test 16 in & 14 in wingspans in relation to final 12 in wingspan

- **More Force Applied**

- Delta wings need high takeoff speeds for optimal results

- **45-Degree Angle of Attack**

- Optimal launch angle for furthest/max distance

References

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